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Epidemiological Study of Hantavirus Infection in the Samara Region of European Russia

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ABSTRACT. European Russia is a highly endemic area of hemorrhagic fever with renal syndrome (HFRS), a rodent-borne zoonotic disease, caused by hantaviruses. In total, 145 small mammals of four species (*Myodes glareolus, Apodemus flavicollis, A. agrarius*, and *A. uralensis*) were trapped in the Samara region of European Russia in August 2005 and examined for the presence of hantavirus (HV). Anti-HV antibodies were found in six of 68 (8.8%) *M. glareolus* and in one of 19 (5.3%) *A. flavicollis* by indirect immunofluorescent antibody assay (IFA). The Puumala virus (PUUV), which is one of the hantavirus species, was detected in the lungs of seven *M. glareolus* by RT-PCR. The virus S-segment was extremely similar (96.2% to 99.3%) to the sequence found in a fatal case of HFRS in the Samara region. Phylogenetic analyses of S and M segments showed that the Samara PUUVs form a cluster within the Russian Volga lineage and apparently differ from other European PUUVs. Anti-PUUV antibodies were found in blood sera from seven HFRS patients and from one undiagnosed patient from the Samara region, using IFA and an enzyme-linked immunosorbent assay (ELISA). These data suggest that the bank vole *M. glareolus* is a primary natural reservoir and vector for PUUV, which is the main causative agent of HFRS in humans in the Samara region.

KEY WORDS: epidemiology, hantavirus, hemorrhagic fever with renal syndrome, Puumala virus, rodents.

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Hantaviruses belong to the genus *Hantavirus*, within the family *Bunyaviridae*, and cause two forms of human illness. These zoonoses include hemorrhagic fever with renal syndrome (HFRS), which occurs in Asia and Europe and is caused by the Hantaan virus (HTNV), Seoul virus (SEOV), Puumala virus (PUUV), Dobrava-Belgrade virus (DOBV), and Amur virus (AMRV). The second form of illness is hantavirus (cardio) pulmonary syndrome (HPS), which occurs in the Americas and is caused by the Sin Nombre virus, Andes virus, and other hantaviruses [13, 28, 33]. Hantaviruses are transmitted by aerosolized excreta from their natural hosts, mainly rodents of the family *Muridae*. Particular hantavirus species are usually harbored by a single rodent species, or by a few closely related rodent species, which indicates co-evolution and co-speciation of hantaviruses with their hosts. The virus genome contains three segments of negative-stranded RNA: the large (L) segment encodes a viral RNA-dependent RNA polymerase, the medium (M) segment encodes a glycoprotein precursor, and the small (S) segment encodes a nucleocapsid protein (NP) [34].

In Russia, five viruses cause HFRS in humans: PUUV and DOBV cause HFRS in European Russia [41], and HTNV, SEOV, and AMRV cause HFRS in Far Eastern Russia [16, 19, 38, 46]. Sporadic HFRS cases, caused by PUUV and DOBV, were recently detected in the western Siberian regions of Russia [45]. The principal hosts for HTNV, AMRV, SEOV, and PUUV are *Apodemus agrarius, A. peninsulae, Rattus norvegicus*, and *Myodes glareolus*, respectively. In Europe, detailed phylogenetic analyses show that the DOBV strains from *A. flavicollis* form a separate evolutionary lineage (DOBV-Af) while strains from *A. agrarius* show higher levels of diversity. Strains from central Europe and central European Russia form the DOBV-Aa lineage, which is distinct from the Saaremaa strains from northeastern Europe [9, 10]. In southern Russia (Sochi district), a new DOBV variant (DOBV-Ap) found in *A. ponticus*, a novel hantavirus natural host, was identified as the causative agent of HFRS [11, 42]. In European Russia and Europe, although the DOBV strains from different *Apodemus* hosts share high amino acid sequence similarity, they can be distinguished in phylogenetic analyses as distinct lineages and seem to possess different virulence in humans as well as in an animal model [12]. The novel DOBV-Ap lineage associated with *A. ponticus*, emerging in an area south of European Russia, confirms the reputation of DOBV as being the most virulent of the European hantaviruses [11]. Rather unusually for hantaviruses, DOBV has already been found in three different Apodemus species. In addition to DOBV,

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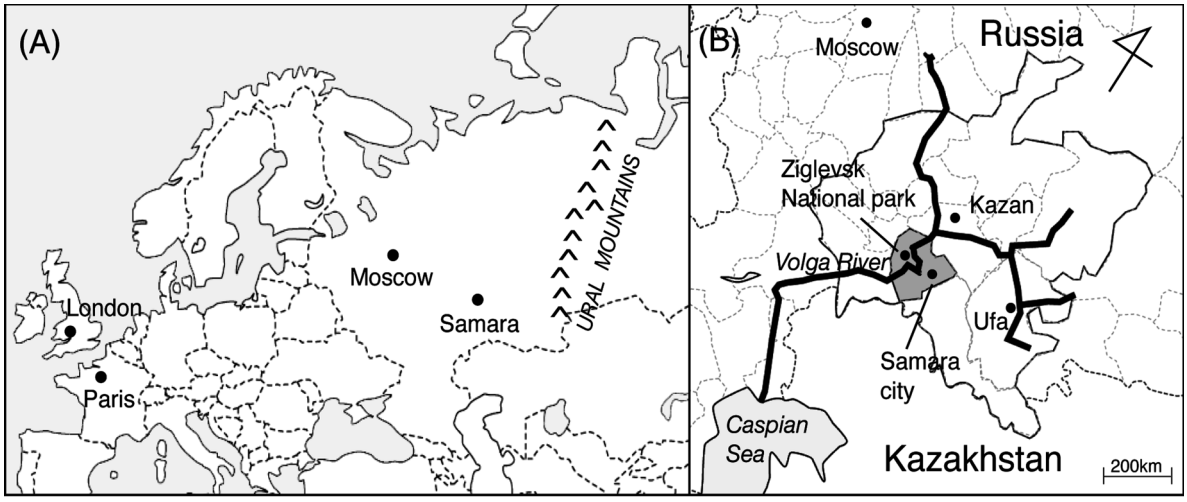


Fig. 1. Map of the survey site. (A) European Russia and European countries. (B) Samara region in the Volga River area; Samara region.

other hantaviruses are also harbored by more than one host species, for example, the Tula virus, which is carried by *Microtus arvalis, M. rossiaemeridionalis*, and *M. agrestis* [29, 35, 36] and SEOV, carried by *Rattus rattus* and *R. norvegicus* [14].

In Russia, HFRS has the highest incidence and morbidity among zoonotic virus infections in humans. The disease has been recognized in Russia since 1930 and has been included in the official reporting system of the Russian Ministry of Public Health since 1978. Of HFRS cases, 97% have been registered in European Russia, while only 3% have been registered in Asian (Far Eastern) Russia [40]. The results of the comparative analyses of clinical data from more than 5,000 HFRS cases from seven different endemic foci, using a common scheme of criteria of severity, indicated that three hantaviruses in Far Eastern Russia (HTNV, SEOV, AMRV) and two hantaviruses in European Russia (PUUV and DOBV) can produce three clinical forms of the disease: mild, moderate, and severe [5, 15]. The rates of severe HFRS caused by HTNV and AMRV in the Far Eastern regions and the DOBV-Ap lineage in southern Russia are significantly higher than the rate of severe HFRS caused by the PUUV, SEOV, and DOB-Aa lineages (Central European Russia). Since HFRS in European Russia is caused by several hantaviruses and the severity varies a great deal, it is inappropriate to use the term "nephropathia epidemica" for the disease. The term HFRS was recommended by the World Health Organization (WHO) Working Group in 1983 to unify the different names of similar clinical diseases in Russia, Europe and Asia [44].

There are approximately 6000 to 8000 clinical cases of HFRS every year, scattered throughout European Russia, that are caused primarily by PUUV. However, the morbidity rates in different administrative regions vary considerably. The highest incidence of HFRS occurs annually in the Ural Mountains and Volga River areas, where there are nine administrative regions with high HFRS morbidity in European Russia [40]. One of the regions in this territory is the Samara region, where we carried out an epizootiological and epidemiological survey in 2005. According to previous studies, most of the HFRS cases in this region seemed to be caused by PUUV infection. However, the characteristics of PUUVs in the region are not well documented and antibodies to HTNV were also detected in healthy blood donors in the region [1]. Therefore, the goals of this study were: (1) to investigate the epidemiological situationy of hantavirus infection both in rodents and humans in the Samara region; (2) to obtain genetic information for hantaviruses, and (3) to identify the epidemiological significance of rodent hosts and hantaviruses with respect to the morbidity of HFRS.

MATERIALS AND METHODS

Rodent survey and human sera: For this study we conducted epizootiological survey targeting rodents. Animals were captured using snap traps that were located in forests near the Zigulevsk National Park, west of the Volga River, and in suburbs of the city of Samara in August 2005 (Fig. 1). Rodent blood samples were collected using filter paper. After complete drying, the filter paper was immersed in 10× the volume of phosphate-buffered saline (PBS) at 4°C overnight. The eluted blood sample was heat-inactivated at 56°C for 30 min and centrifuged at 300 × g for 5 min. Thereafter, the supernatant was used as 1:10 diluted serum. The lungs, hearts, livers, kidneys, and spleens were collected from the captured rodents and stored at –80°C until use. Human blood sera were obtained from clinically diagnosed HFRS patients and patients with high fever of unknown etiologies. Blood sera were stored at –80°C until use.

Indirect immunofluorescent antibody assay (IFA): Anti-hantavirus antibodies in rodents and patients were detected

Table 1. Primers used for RT-PCR and sequencing of S and M genome segments of hantaviruses

Virus	Gene	Primer name	Primer sequence	Position
Puumala	S	sotkamo62Fw	tccaagaggatataacccgccat	62–84
		sotS172Fw	ctgcaagccaggcaacaaacagtgtcagca	172–201
		sotS593Fw	ctcagtccaccatgaag	593–609
		ufa97S850Fw	aagccagaagttaaacct	850–867
		samS1146Fw	atatttgcgccggacacaatc	1146–1166
		samS1286Fw	gtcactcatcgatcagaagg	1286–1305
		samS1552Fw	cagggaattactaatgacag	1552–1571
		sotS537Rv	ctcaaatgatgtgtcatcc	537–519
		sotS894Rv	gtctgccacatgatttttgtcaagcacatc	894–865
		samS1037Rv	aaytcagccatcccagcaac	1037–1018
		samS1500Rv	gataataataattgtcaaacccg	1500–1478
		PUU1801SRv	atcagcatgttgaggtagta	1801–1782
	M	Ufa97M1789F	tctcaatccatctgaggcaacaac	1789–1812
		sam94M2122f	gctatacatataggagacag	2122–2141
		sam94M2546F	actacttcagtcaaagtgtg	2546–2565
		sam94M2956F	gatttaagtgaaacaccatgcca	2956–2978
		sam94M3303F	tgatgatggtgcaccagag	3303–3321
		sam94M2019R	gtctcagcactagcagcccatac	2019–1997
		sam94M2317R	gcagtctgccaaggataagc	2317–2298
		sam94M2478R	ctggtataccttaaggacac	2478–2459
		sam94M2885R	ctaatgcacttgcagatatg	2885–2866
		sam94M3315R	tggtgcaccatcatcaaag	3315–3297
		Ufa97M3654R	ccaggcataatcggtatggggtaa	3654–3631
Dobrava	S	DOBS84Fw	caattggtgatagccaggcagaagg	84–108
		DOBS1012Rv	gccatgcctgcaatgaacaggcagg	1012–988

by IFA. Antigen slides were prepared using Vero E6 cells (ATCC No. CRL-1586) separately infected with the PUUV strain Sotkamo, the HTNV strain 76–118, and the SEOV strain SR-11. Serially diluted rodent blood sera (from 1:16 to 1:2048) were applied to the antigen slides which were then incubated at 37°C for 1 hr. After three washes with PBS, Protein G Alexa Fluor® 488 conjugate (Invitrogen, Carlsbad, CA, U.S.A.) or Alexa Fluor® 488 Goat Anti-Mouse IgG (Invitrogen) was spotted on to the slides. After incubation at 37°C for 1 hr, the slides were washed and observed by fluorescence microscopy. Scattered granular fluorescence in the cytoplasm of Vero E6 cells was considered as a positive reaction.

Enzyme-linked immunosorbent assay (ELISA): Anti-hantavirus IgG in patients was detected by ELISA [2, 20, 27]. Briefly, 96-well plates were coated with the monoclonal antibody (MAb) E5/G6 [26, 47] as a capture antibody and were treated with Block Ace (Dainippon Sumitomo Pharma Co., Ltd., Osaka, Japan) at 37°C for 1 hr for blocking. The Baculovirus-expressed recombinant hantavirus NP of PUUV, DOBV, or HTNV [2, 20] was then added to the plates and incubated at room temperature for 1 hr. Sera from clinically diagnosed HFRS patients were diluted to 1:200, added to the plates, and incubated at room temperature for 1 hr. Alkaline phosphatase-conjugated anti-human IgG (Sigma-Aldrich Cooperation, St. Louis, MO) was diluted to 0.5 μg/m*l*, added to the plates, and incubated at room temperature for 1 hr. Finally, p-nitrophenyl phosphate solution (Sigma-Aldrich) was added to the plates, which were incubated at room temperature for 1 hr, and absorbance values at 405 nm (A405) and 650 nm (A650) were measured using a spectrophotometer. The IgG-ELISA value was expressed as the value of A405 minus A650. Between the steps of the ELISA assay, the plates were washed three times with PBS containing 0.05% Tween 20 (PBST).

Anti-hantavirus IgM in patient sera was detected by μ-capture ELISA. Briefly, 96-well plates were coated with goat-anti human IgM (MP Biomedicals United States, Solon, OH) as the capture antibody. After blocking, 1:200 diluted patient sera were added to the plates and incubated at room temperature for 1 hr. Recombinant NP proteins of PUUV, DOBV, and HTNV were then added to the plates and incubated at room temperature for 1 hr. Biotinylated E5/G6 mAb and peroxidase-conjugated streptavidin (Sigma-Aldrich) were applied consecutively and incubated at room temperature for 1 hr. Thereafter, 3,3',5,5'-tetramethylbenzidine solution (Sigma-Aldrich) was added to the plates as a substrate. The colorimetric reaction was stopped by adding 50 μ*l* of 0.5 M H_2SO_4. The IgM-ELISA value was calculated as described for the IgG-ELISA.

Reverse transcription-polymerase chain reaction (RT-PCR): Total RNA from the lung tissue of *M. glareolus* and *A. flavicollis* was extracted using ISOGEN (Nippon Gene Co., Ltd., Osaka, Japan). The RNA (5 μg) was reverse-transcribed using 200 units of Superscript II RNase H-reverse transcriptase (Invitrogen Corporation) and 500 *n*g random primers (Invitrogen) according to the manufacturer's

instructions. The S and M hantavirus segments were amplified by PCR using Platinum® Taq DNA polymerase High Fidelity (Invitrogen). The thermal conditions for PCR were 94°C for 2 min, followed by 35 cycles of 94°C for 30 sec, 50°C to 60°C (depending on the primers used) for 30 sec, and 68°C for 2 min. The primers used for PCR are listed in Table 1. Primers SotS172Fw and SotS894Rv were used to detect PUUV, and DOBS84Fw and DOBS1012Rv were used to detect DOBV (Table 1).

Sequencing of hantavirus genome segments: The amplified DNA fragments were separated using the Wizard® SV Gel and PCR Clean-Up System (Promega, Madison, WI). The purified DNA fragments were sequenced using the Big Dye Terminator v1.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, CA, U.S.A.) according to the manufacturer's instructions. Primers used for amplification and sequencing are listed in Table 1.

Phylogenic analysis: The ClustalX version 1.83 (available from the URL: ftp://ftp.ebi.ac.uk/pub/software/clustalw2) was used to generate the alignments, and phylogenetic trees were formed by the neighbor-joining method [39]. The reliability of the dendrogram was evaluated by 1,000 bootstrap replicates.

RESULTS

Rodent survey: We carried out an epizootiological survey and captured 145 rodents in the Samara region. The rodent species included 68 *M. glareolus*, 19 *A. flavicollis*, 21 *A. agrarius*, and 37 *A. uralensis*. Six *M. glareolus* (8.8%) and one *A. flavicollis* (5.3%) harbored anti-hantavirus antibodies (Table 2). No antibodies were detected in *A. agrarius* or *A. uralensis*. Antibody titers to PUUV, measured by IFA, in *M. glareolus* ranged from 1:256 to 1:1024, and showed titers more than two times higher than antibody titers to HTNV or SEOV. One *A. flavicollis* had antibodies to both HTNV and SEOV at 1:16 (Table 3) but was negative for PUUV. Total RNA samples extracted from the lungs of 68 *M. glareolus* and 19 *A. flavicollis* were subjected to RT-PCR. Hantavirus genes were detected in four seropositive, and three seronegative *M. glareolus* (Tables 2 and 3), whereas no hantavirus genes were detected in *A. flavicollis*.

Sequencing of hantavirus genes: Partial S- and M-segments were sequenced from RT-PCR-positive *M. glareolus*. The viruses carried by *M. glareolus* (animal numbers: 6, 49, 68, 94, 112, 128 and 147) were named Samara_6/CG/2005 (Sam6); Samara_49/CG/2005 (Sam49); Samara_68/CG/2005 (Sam68); Samara_94/CG/2005 (Sam94); Samara_112/CG/2005 (Sam112); Samara_128/CG/2005 (Sam128) and Samara_147/CG/2005 (Sam147), respectively. The sequences determined and used in this study are listed in Table 4.

The S-segment sequences of viruses in the Samara region, which were similar to those of the PUUV strains Kazan, CG1820, and DTK/Ufa-97, identified near the Samara region within the Volga river area, demonstrated

Table 2. Detection of anti-hantavirus antibodies and virus RNA among captured rodents in the Samara region

Rodent species	Number of rodents	Seroprevalence (%)	Detection rate of Virus RNA (%)[a)]
M. glareolus	68	6/68 (8.8)	7/68 (10.1)
A. flavicollis	19	1/19 (5.3)	0/19 (0.0)
A. agrarius	21	0/21 (0)	ND[b)]
A. uralensis	37	0/37 (0)	ND
Total	145	7/145 (4.8)	

a) Virus RNA was detected from rodent lungs by RT-PCR.
b) ND: not done.

Table 3. Rodents with anti-hantavirus antibodies and virus RNA

Rodent species	Point of capture	Animal No.	IFA titer			Virus RNA[a)]
			PUUV	HTNV	SEOV	RT-PCR
M. glareolus	Suburb of Samara	66	256	<16	<16	–[b)]
		137	512	256	128	–
		68	1024	<16	<16	+[c)]
		147	1024	64	64	+
		94	<16	<16	<16	+
	Zigulevsk national park	6	<16	<16	<16	+
		49	<16	<16	<16	+
		112	512	64	64	+
		128	256	<16	<16	+
A. flavicollis	Suburb of Samara	59	<16	16	16	–

a) Virus RNA was detected from rodent lungs by RT-PCR. b) negative. c) positive.

Table 4. Hantavirus sequences used in this study

Virus name	Strain	Source	Country/ Region	Accession No.	
				S segment	M segment
Hantaan	76–118	*Apodemus*	Korea	M14626	M14627
Tula	Moravia/ 5302v/95	*Microtus arvalis*	Czech Republic	Z69991	Z69993
Khabarovsk	Ls136V	*Microtus fortis*	Russia	AJ011646	
Hokkaido	Kamiiso	*Myodes rufocanus*	Japan	AB010730	
	Tobetsu	*M.rufocanus*	Japan	AB010731	
Puumala	Balkan-1	*Myodes glareolus*	Balkan	AJ314600	
	Balkan-2	*M. glareolus*	Balkan	AJ314601	
	Klippitztoerl	*M. glareolus*	Austria	AJ888751	
	Ernstbrunn	*M. glareolus*	Austria	AJ888752	
	Opina	*M. glareolus*	Slovakia	AF294652	
	Mignovillard	*M. glareolus*	France	AM695638	
	CG13891	*M. glareolus*	Belgium	U22423	U22418
	Cg-Erft	*M. glareolus*	Belgium	AJ238779	AJ238778
	Thuin	*M. glareolus*	Belgium	AJ277030	
	Couvin	*M. glareolus*	Belgium	AJ277034	AJ277040
	Eidsvoll	*M. glareolus*	Norway	AJ223368	
		M. glareolus	Norway	AJ223369	
	Solleftea	*M. glareolus*	Norway	AJ223376	
		M. glareolus	Norway	AJ223377	
	Fin	*M. glareolus*	Denmark	AJ238791	
		M. glareolus	Denmark	AJ278092	
		M. glareolus	Denmark	AJ278093	
	Umea/hu	HFRS patient	Sweden	AY526219	AY526218
	Vindeln	*M. glareolus*	Sweden	Z48586	Z49214
	Vranica	*M. glareolus*	Sweden	U14137	U14136
	Virrat	*M. glareolus*	Finland	Z69985	
	Evo	*M. glareolus*	Finland	Z30702	
		M. glareolus	Finland	Z30703	
	Pallasjarvi	*M. glareolus*	Finland	AJ314597	
	Kolodozero	*M. glareolus*	Finland	AJ238789	
	Karhumaki	*M. glareolus*	Finland	AJ238788	
	Gomselga	*M. glareolus*	Finland	AJ238790	
	Sotkamo	*M. glareolus*	Finland	X61035	X61034
	1324Cg/79	*M. glareolus*	Finland	Z46942	

94% to 95% nucleotide similarity identity and 98% similarity identity in the deduced amino acid sequences (Table 5). The homology identities of the nucleotide and amino acid sequences between the viruses in the Samara and Scandinavian PUUVs (strains Sotkamo and Umea/Hu) were 85% and 96%, respectively.

In addition, comparisons of S-segments between sequences identified from the Zigulevsk National Park (Sam6, Sam49, Sam112, and Sam128) and F-s808, which was detected from a fatal case of HFRS in the Samara region, showed 99.3% and 100.0% homology at the nucleotide and amino acid levels, respectively (Table 5).

Phylogenetic analysis: Phylogenetic analyses of hantavirus S- and M-segments were performed (Fig. 2). The result clearly indicated that the hantaviruses detected in *M. glareolus* belong to the Volga River area lineage of PUUV. This lineage consists of viruses originating from the Samara region (Sam6, Sam49, Sam68, Sam94, Sam112, Sam128, Sam147, F-s808); Tatarstan (Kazan); Bashkortostan (CG17/Bashkiria-2001, CG1820, K27, DTK/Ufa-97); Saratov (P360) and Udmurt (Udmurtia/894Cg/91, Udmurtia/444Cg/88). They occupied one cluster and were distant from the viruses derived from other Russian regions (such as Omsk) and European countries (Fig. 2). The virus clusters were formed according to the geographical origin of the viruses.

Serological analysis of patient sera: To examine the causative agents of HFRS, serological analyses were performed on 12 blood sera from seven patients from the Samara region who were clinically diagnosed with HFRS, as well as on nine sera from six patients with high fevers of unknown etiologies. Reactivities of sera to PUUV, SEOV, and HTNV were analyzed by IFA (Table 6). Sera from six HFRS patients (Patients 1, 2, 3, 4, 6, and 7) and from one of the unknown fever patients (Patient 11) contained anti-hantavirus antibodies. In all positive sera, IFA titers to PUUV were at least 16 times higher than titers to SEOV and HTNV. In patient numbers 1, 2 and 3, titers of convalescent sera to PUUV were higher than those of acute phase sera

Table 4. continued. Hantavirus sequences used in this study

Virus name	Strain	Source	Country/ Oblast/Region	Accession No.	
				S segment	M segment
Puumala	CG144	*M. glareolus*	Russia/ Omsk	AF367064	
	CG215	*M. glareolus*	Russia/ Omsk	AF367066	
	CG222	*M. glareolus*	Russia/ Omsk	AF360067	
	Baltic	*M. glareolus*	Baltic	AJ314598	
		M. glareolus	Baltic	AJ314599	
	CG17	*M. glareolus*	Russia/ Bashkortostan	AF442613	AF442614
	CG1820	*M. glareolus*	Russia/ Bashkortostan	M32750	M29979
	K27		Russia/ Bashkortostan	L08804	L08754
	P360		Russia/ Bashkortostan	L11347	L08755
	DTK/Ufa-97	HFRS patient	Russia/ Bashkortostan	AB297665	AB297666
	Udmurtia	*M. glareolus*	Russia/Udmurt	Z21497	
		M. glareolus	Russia/Udmurt	Z30706	
	Kazan	*M. glareolus*	Russia/ Tatarstan	Z84204	Z84205
	Samara_6/ CG/2005	*M. glareolus*	Russia/Samara Zigulevsk	AB433842	AB433849
	Samara_49/ CG/2005	*M. glareolus*	Russia/Samara Zigulevsk	AB433843	AB433850
	Samara_112/CG/2005	*M. glareolus*	Russia/Samara Zigulevsk	AB433846	AB433853
	Samara_128/CG/2005	*M. glareolus*	Russia/Samara Zigulevsk	AB433845	AB433852
	Samara_68/ CG/2005	*M. glareolus*	Russia/Samara city	AB433844	AB433851
	Samara_94/ CG/2005	*M. glareolus*	Russia/Samara city	AB433847	AB433854
	Samara_147/CG/2005	*M. glareolus*	Russia/Samara city	AB433848	AB433855
	F-s808	HFRS patient	Russia/Samara city	AF411446	

Table 5. Comparison of nucleotide (nt 85 to 1238) and amino acid (aa 15 to 403) sequences of the S segment among hantaviruses

	Identities of nucleotide and amino acid (%)[a]											
	Puumala										Dobrava	Hantaan
	Zigulevsk[b]	Sam68[c]	Sam94[d]	Sam147[e]	F-s808	Kazan	CG1820	Ufa97	Sotkamo	Umea/hu		76–118
Zigulevsk	–	96.5	96.7	96	99.3[f]	95.1	94.2	94.3	85.5	84.5	62.8	62.2
Sam68	99.7	–	98.7	99.5	96.7	94.8	93.5	93.6	85.5	84.6	62.6	62.8
Sam94	99.3	99.2	–	98.2	96.7	94.9	93.5	93.6	85.6	84.4	63.4	62.3
Sam147	98.7	99	98.2	–	96.2	95.2	94.1	94.2	85.2	84.2	62.6	62.9
F-s808	100	99.7	99.3	98.7	–	95.4	94.3	94.4	85.7	84.9	61.8	61.7
Kazan	99.1	98.7	98.4	97.7	99	–	94.3	94.4	85.3	84.2	63	62.4
CG1820	99.1	98.7	98.4	97.7	99	98.6	–	99.8	85.6	84	62.7	62.3
Ufa97	99.3	98.7	98.6	97.7	99.3	98.8	99.3	–	85.6	84.1	62.6	62.2
Sotkamo	96.8	96.7	96.1	96.4	96.8	96.8	96.3	96.5	–	85.5	63.4	62.7
Umea/hu	96.3	96.4	95.6	95.9	96.5	96.3	95.8	96.1	96.5	–	63.2	61.3
Dobrava	61	61.6	60.3	60.6	58.6	60.5	60.5	60.5	61.2	60.7	–	74
Hantaan	61.4	60.2	60.7	59.6	59.3	61.2	61	61	60.7	60.7	83	–

a) Values to the right above the diagonal show nucleotide identities; those to the left below the diagonal show amino acid identities.
b) Zigulevsk; Samara_6/CG/2005, Samara_49/CG/2005, Samara_112/CG/2005 and Samara_128/CG/2005 (the same sequence).
c) Samara_68/CG/2005.
d) Samara_94/CG/2005.
e) Samara_147/CG/2005.
f) Values in bold show the identities between the F-s808 and PUUVs detected from *M. glareolus* in Zigulevsk National Park

(Table 6). However, no PUUV genes were detected in the sera of HFRS patients by RT-PCR (Table 6). ELISA was also used to perform serological analysis using baculovirus-expressed NPs of HTNV, DOBV, and PUUV. All IFA-positive sera had at least two times higher IgG-ELISA values to PUUV than to DOBV or HTNV (Fig. 3A). IgM-ELISA values of the patients also showed the same reactive patterns as IgG-ELISA values (Fig. 3B).

DISCUSSION

The purpose of this study was to investigate the characteristics and ecology of hantaviruses in European Russia. There is an urgent need for clear information concerning the circulation and maintenance of hantaviruses in nature, their association with specific rodent reservoir hosts, and their contribution to HFRS morbidity. The majority of HFRS cases in European Russia are caused by PUUV, with only rare cases caused by DOBV. Human epidemics are charac-

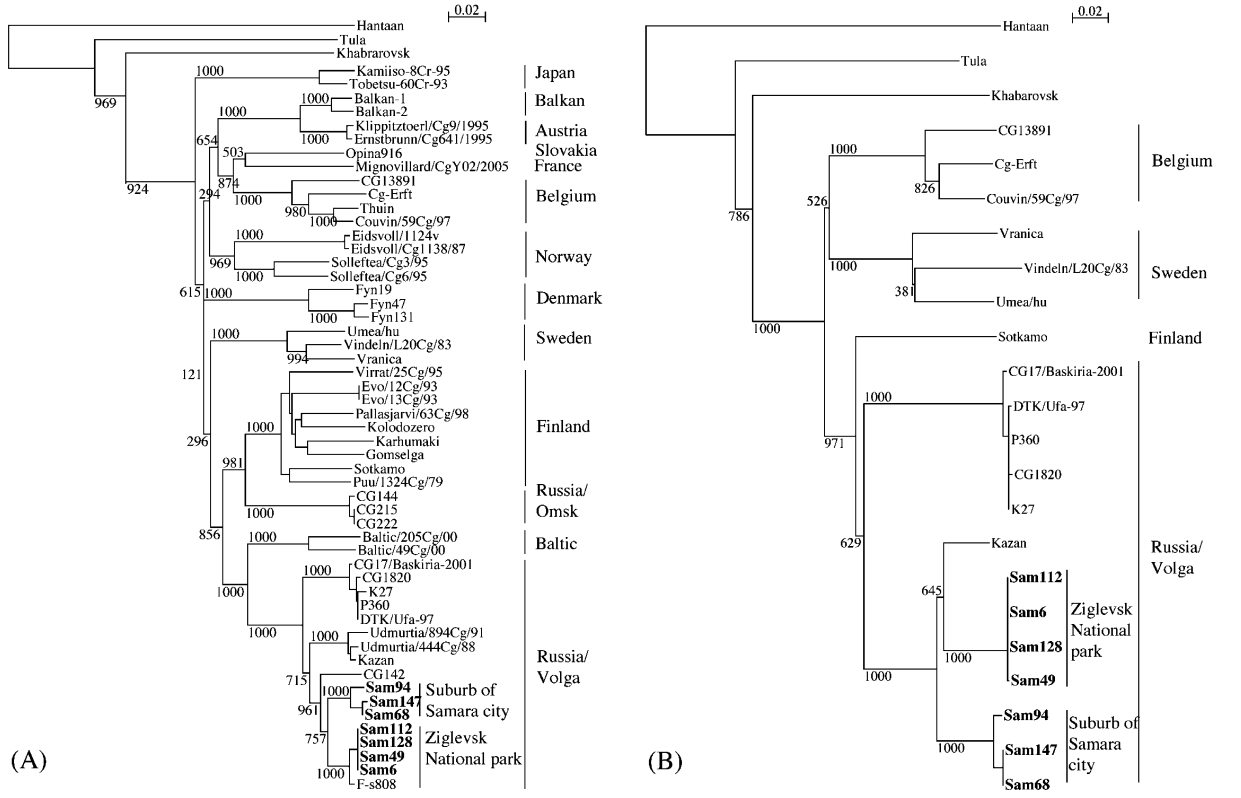


Fig. 2. Hantavirus phylogenetic trees. Hantavirus sequences were aligned using Clustal X, and the phylogenetic trees were generated using the neighbor-joining method. The reliability of the dendrogram was evaluated by 1,000 bootstrap replicates. (A) Phylogenetic tree of hantavirus partial S-segments (nts 85 to 1238). (B) Phylogenetic tree of hantavirus partial M-segments (nts 2484 to 3494).

terized by outbreaks with a frequency of three to four years. The primary determinative factor that influences HFRS epidemics in humans is the periodical and massive reproduction of rodents, together with the development of epizootics in their expanding populations [40].

The results of our study show that PUUV is the major causative agent of HFRS in the Samara region and that *M. glareolus* plays an important role as the reservoir animal in this area. In addition, anti-hantavirus antibodies were detected in one *A. flavicollis* with low titers to HTNV and SEOV but not to PUUV. However, no viral genes were detected in *A. flavicollis*. Further investigation is needed to determine whether DOBV or DOB-related viruses circulate among the *A. flavicollis* population in the Samara region.

Although serum samples were limited, the serological analyses of HFRS patients suggested that PUUV is the major cause of HFRS cases in the Samara region. This observation strongly supports previous findings [21, 25, 43]. The virus sequences of Sam6, Sam49, Sam112, and Sam128 identified in *M. glareolus* at Zigulevsk National Park were almost identical to PUUV from a fatal case of HFRS that occurred in the Samara region. Therefore, it is highly possible that *M. glareolus* in Samara carries PUUV, which causes severe HFRS. Further characterization of the viruses in the Samara region is required to elucidate the pathogenicity of PUUV.

Nucleotide sequence comparisons revealed that the viruses in the Samara region are similar to the PUUV strains Kazan, CG1820, and DTK/Ufa-97 that were isolated near the Samara region within the Volga River area. In addition, phylogenetic analysis clearly demonstrated the presence of PUUVs in the Samara region; viruses originating from the Volga River area occupied one cluster and were distant from European PUUVs. Viruses of Russian origin formed several separate lineages such as those found in Omsk and in the Volga River area. These findings provide insight towards understanding the evolution of PUUV [4, 22, 37].

PUUV is the most common cause of HFRS in Europe and is harbored by the bank vole (*M. glareolus*), which is found in most of Europe. Genetic analyses of strains from the Baltic countries, Austria, Slovakia, France, Belgium, Norway, Denmark, Sweden, Finland, Germany, and Russia have shown PUUV to be the most variable of the hantavirus species; genetic diversity at the nucleotide level reaches 20% and 17% for the coding regions of the M- and S-segments, respectively, and reaches even higher values (37% and 30%) for their 3'-noncoding regions [6, 8, 17, 18, 29, 32].

Interestingly, PUUV strains from the Omsk region (west-

Table 6. Detection of anti-hantavirus antibodies in HFRS patients and patients with high fever of unknown etiology

Patient No.	Serum No.	Age	Diagnosis	Days after onset	IFA titer PUUV	IFA titer SEOV	IFA titer HTNV	PCR
1	1–1	43	HFRS	6	256	<32	<32	–[a)]
	1–2			20	512	<32	<32	ND[b)]
2	2–1	58	HFRS	14	1024	<32	<32	–
	2–2			19	2048≤	<32	<32	ND
3	3–1	44	HFRS	6	512	<32	<32	–
	3–2			26	2048≤	<32	<32	ND
4	4–1	16	HFRS	7	256	<32	<32	–
	4–2			10	256	<32	<32	ND
5	5–1	18	HFRS	1	<32	<32	<32	–
	5–2			6	<32	<32	<32	ND
6	6–1	29	HFRS	10	1024	<32	<32	–
7	7–1	38	HFRS	8	2048≤	<32	<32	–
8	8–1	44	Unknown	4	<32	<32	<32	–
	8–2			12	<32	<32	<32	ND
9	9–1	21	Unknown	5	<32	<32	<32	–
	9–2			8	<32	<32	<32	ND
10	10–1	52	Unknown	8	<32	<32	<32	–
	10–2			11	<32	<32	<32	ND
11	11–1	28	Unknown	7	256	<32	<32	–
12	12–1	28	Unknown	Unknown	<32	<32	<32	–
13	13–1	35	Unknown	5	<32	<32	<32	–

a) –: negative.
b) ND: not done

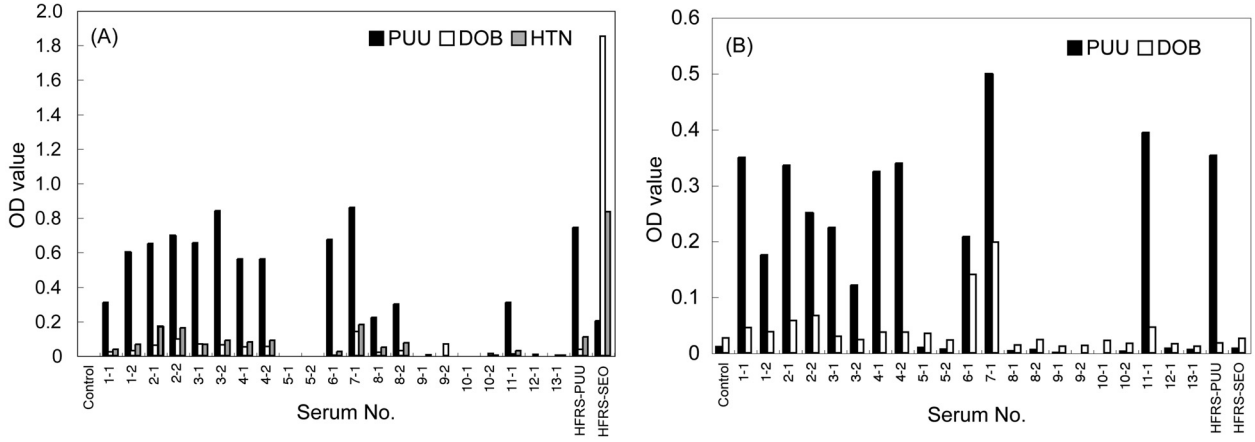


Fig. 3. Detection of anti-hantavirus antibodies in HFRS patients and patients with high fevers of unknown etiologies by ELISA. Baculovirus-expressed hantavirus nucleocapsid proteins (NPs) were used as the antigens. PUU: full length NP of PUUV; DOB: full length NP of DOBV; HTN: full length NP of HTNV; Control: Normal human serum; HFRS-PUU: PUUV-infected HFRS patient serum; HFRS-SEO: SEOV-infected HFRS patient serum. (A) Detection of anti-hantavirus IgG. (B) Detection of anti-hantavirus IgM.

ern Siberia) are more closely related to Finish strains than to the Russian strains isolated from geographically closer regions (e.g., Ufa, Kazan, Udmurtia, or Samara) [4]. Closely related sublineages of PUUV were found in samples obtained from near and distant geographical locations of Russia [4, 7, 23, 24,]. According to a theory proposed by

Plyusnin *et al.* [30,31], the extent of hantavirus diversity is proportional to the geographic distance between the areas of their circulation. However, while the microgeographical relationships among hantaviruses might be linearly proportional, the macrogeographical relationships are likely to be more complex. The macrogeographical evolution of hantaviruses might be significantly biased by distant migrations of animals, especially ones that occurred during the several interglacial epochs [3, 4].

The results obtained in this study further characterize the public health threat caused by PUUV infection in the high HFRS epidemic area of European Russia. The information obtained in this study is quite valuable to prevent HFRS by avoiding contacts to *M. glareolus* and is also quite important to understand the significance of the vaccine development to PUUV infections in European Russia.

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